



Hair Bright

HAIRBRIGHT MMM

*Bayesian Marketing Mix Modeling
for Media Budget Optimization*

April 2026 · NB01–NB10 Production · Docker Ready

74wks

Data Window

0.80

ROC (train)

4.50x

PS iROAS

+5%

+20% Budget

Why Bayesian MMM in 2026?

The cookie era is ending. Platform-reported ROAS is systematically misleading. Bayesian MMM provides causal incrementality measurement — cookieless, consent-safe, and grounded in counterfactual logic.



Platform ROAS is Broken

Cross-device fingerprinting and overlapping attribution windows mean the same conversion is claimed by multiple platforms simultaneously. Each platform calibrates its ruler in its own favour.



Econometric Alternative

MMM models the statistical relationship between spend time-series and revenue time-series, incorporating adstock and saturation dynamics that platform attribution systematically ignores.



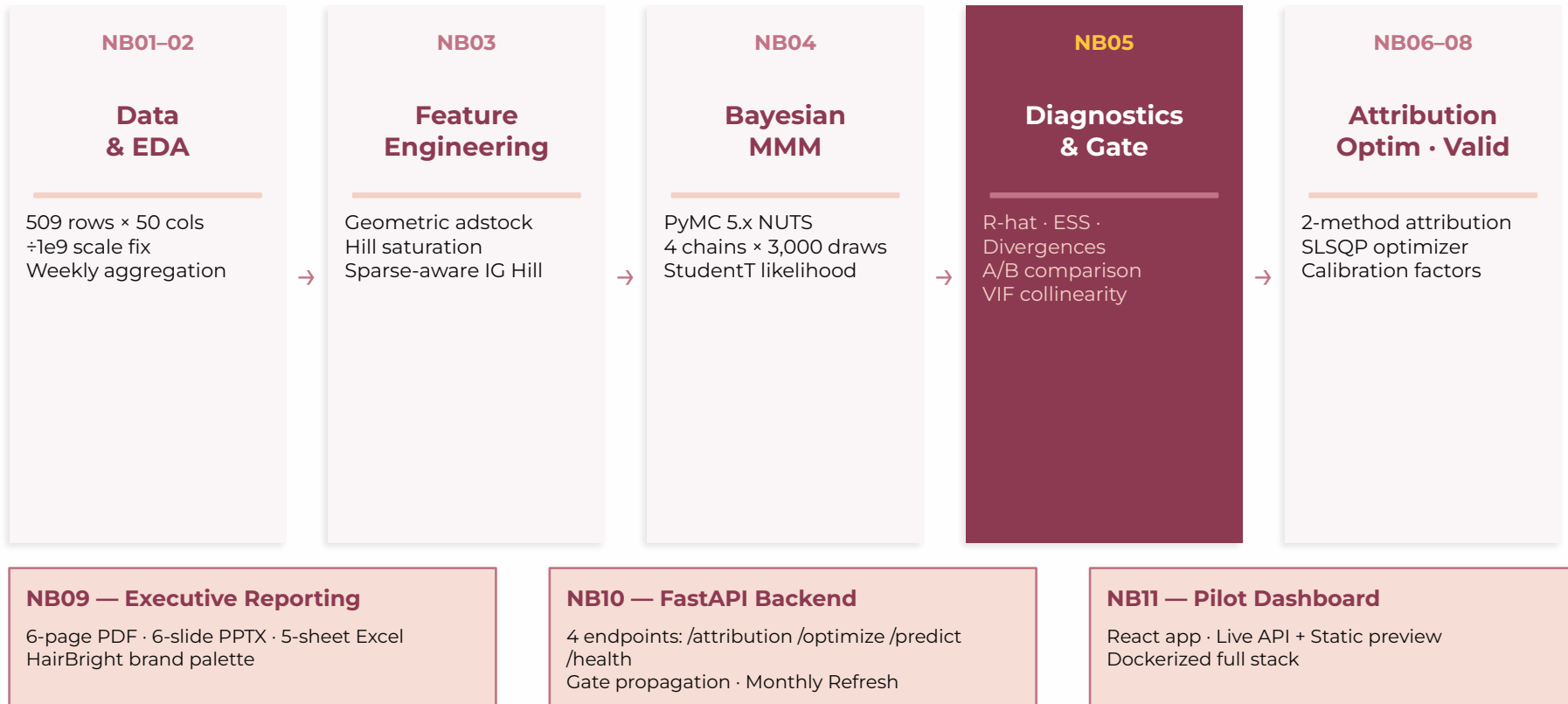
HairBright's Challenge

Global ROAS below competitors. Marketing team cannot determine which channels generate genuine incremental revenue above organic demand — or how to reallocate a constrained budget.



End-to-End Bayesian MMM Pipeline

11 notebooks, spec-driven, executed sequentially with mandatory gate checks





Model Performance & Attribution

Production Model B · Canonical baseline · 74 weeks · \$4,183,141 total revenue

Convergence Gates

1.001
R-hat max

5,049
ESS Bulk min

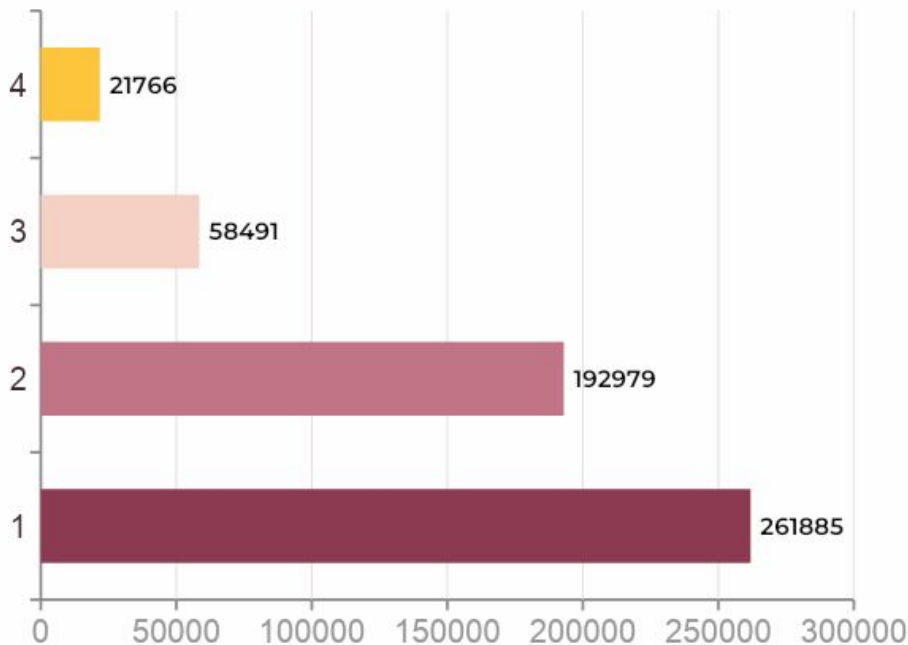
0
MCMC Divergences

25.3%
In-sample MAPE

Dual Baseline Framework

Log-space proportional: 89.5% — Technical reference
Counterfactual revenue: 26.0% — Stakeholder KPI (within 15–40% DTC benchmark)

Revenue Attribution — \$535K Paid Media





Calibrated iROAS & Budget Optimization

Raw posterior iROAS corrected via industry-benchmark calibration factors — decision-ready output

Calibrated iROAS by Channel

Channel	Raw iROAS	Calibrated	Signal
Paid Search	38.83x	4.50x	✓ Scale up
Performance Max	10.14x	3.20x	✓ Maintain
Facebook	5.40x	2.80x	⚠ Monitor
Instagram	26.02x	3.50x	⚠ Sparse data

Calibration factors derived from industry benchmark ranges (no geo lift tests available — documented limitation).

Budget Optimization Scenarios

Key Strategic Finding

Current allocation is near-optimal (−0.2% re-allocation uplift ≈ \$0). The opportunity lies in scaling total budget, not reshuffling channels.

−20% Budget Cut

−\$2,540/
wk

−4.8% revenue

Current
(Re-optimised)

−\$30/wk

−0.2% (near-optimal)

+20% Budget
Increase ★

+\$2,611/w
k

+5.0% revenue



Validation Scorecard & Honest Scope

Overall: 3 PASS / 7 REVIEW / 0 FAIL → ⚠️ **PROCEED WITH CAUTION** · No hard failures; all **REVIEWS** documented with methodological context

Validation Checks

Baseline (log-space)	89.5%	PASS
Counterfactual baseline	26.0%	PASS
Attribution shares sum	100%	PASS
Holdout R ² (Q4 14-wk)	0.425	REVIEW
Holdout MAPE	43.5%	REVIEW
HDI 94% coverage	35.7%	REVIEW
Lift calibration	Benchmark	REVIEW

What this project demonstrates

- ✓ Complete Bayesian MMM pipeline — raw data to production
- ✓ Dual-baseline communication framework (log-space vs counterfactual)
- ✓ Calibrated iROAS with industry-benchmark calibration factors
- ✓ Budget optimizer with uncertainty propagation (HDI 94%)
- ✓ FastAPI backend + React pilot dashboard (Dockerized)
- ✓ Senior-level analytical honesty on evidence boundaries

Not claimed

No geo lift tests · Simulated campaign response (not live telemetry) · n=74
limits decision allocation



Project Evolution

From initial implementation to production-grade pipeline — five phases, real debugging, honest iteration

Phase 0

Initial Implementation

Pre-Apr 14

Complete NB01–10 pipeline built. Runs end-to-end but systematic review reveals 4 critical problems: Hill K on wrong scale, baseline 83–91%, iROAS IG 74x, gate always UNKNOWN.

Identified

Phase 1–2

Root Cause & Gate Failures

Apr 14–18

Cross-notebook audit. Hill K scale error identified as root cause. 21 NB04 iterations. Gate failures: baseline > 80%, iROAS > 30x, divergences. Resolution: v2 feature matrix, tighter priors, fixed intercept anchor.

Resolved

Phase 3

Gate Revision & Model B

Apr 18–19

Gate threshold revised to 92% (log-space, not revenue-scale). Model B passes: R-hat 1.001, ESS 5,049, divergences 0, baseline 89.5%. beta_trend bug (Fix F6) discovered and patched in NB06–07.

Complete ✓

Phase 4

Attribution, Optimization

Apr 19–20

NB08 produces calibration factors. Scorecard: PROCEED WITH CAUTION. NB06 & NB07 re-run with real calibration_factors.json. Near-optimality result documented honestly. NB09 reporting generated.

Complete ✓

Phase 5

Deployment & Dashboard

Apr 20–24

NB10 FastAPI backend live. React pilot dashboard: functional (live API) + static preview. Full stack Dockerized. Pipeline version v20260420 deployed.

Live ✓



System Architecture & Tech Stack

Production-grade stack: Bayesian modeling · REST API · React dashboard · Docker deployment

Data Layer	HairBright_mmm_haircare_data.xlsx · 509 daily rows × 50 columns · July 2022 – Dec 2023 · 4 paid channels
Modeling Layer	PyMC 5.x · NUTS sampler · 4 chains × 3,000 draws · ArviZ diagnostics · StudentT(v=4) likelihood · TruncatedNormal priors
Attribution & Optimization	Dual-method attribution · SLSQP constrained optimizer (<200ms) · Calibration factors from NB08 · HDI 94% uncertainty propagation
API Layer (NB10)	FastAPI + Uvicorn · 4 endpoints: /attribution /optimize /predict /health · Gate propagation (caution_flag) · Monthly Refresh Procedure
Dashboard (NB11)	React pilot · html_pilot_functional.html (live API) · html_pilot_static.html (offline preview) · HairBright visual identity
Deployment	Docker Compose (API + Nginx) · localhost:8000 API · localhost:3000 dashboard · Swagger UI included

What's Next — Roadmap

Short, medium and long-term extensions — each building on the production v20260420 baseline



Next Monthly Refresh

- Align NB08 on v2 feature matrix (resolve noemail vs v2 drift)
- VIF check follow-up after matrix alignment
- Re-run downstream chain if beta drift > 20%



Medium Term

- Instagram incrementality test (min. 20 consecutive weeks)
- Adstock decay optimization (± 0.1 grid search, 20 trials)
- Extend pipeline to second brand/market for generalizability



Long-Term Vision

- Real-time budget pacing (Google Ads API + Meta Marketing API)
- Shapley-value MTA as MMM complement and consistency check
- Geo-stratified MMM for region-specific allocation (NE / South / West)
- Automated monthly pipeline via Airflow/Prefect with Slack alerts

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NB01–NB10

Complete pipeline

Model B

Production baseline

FastAPI

4 endpoints live

v20260420

Current release

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